

Presentation of Impact Sensations by a Floor-Mounted Turntable

Mutsuki Shimada*

The University of Electro-Communications

Izumi Mizoguchi†

The University of Electro-Communications

Hiroyuki Kajimoto‡

The University of Electro-Communications

ABSTRACT

Previous studies on impact sensation presentation have primarily focused on delivering impulses to the hands using wearable or handheld devices. On the other hand, impact reaction forces propagate through the entire body, ultimately acting between the feet and the ground. Leveraging this physical property, we hypothesize that presenting reaction forces to the feet can enhance impact sensation. This study investigates a method using a floor-mounted turntable. As the user stands on the device, it rotates impulsively during a striking motion to present a reaction force, thereby enhancing the realism of the impact experience.

Index terms: Force Feedback, Haptics, Impact Sensation, Reaction Force.

1 INTRODUCTION

To enhance the sense of presence in VR experiences, the appropriate presentation of impact sensation—the haptic feedback upon collision—is crucial. Impact sensation comprises both vibration (high-frequency components) and reaction force (including direct current and low-frequency components) [3]. Conventional research has primarily focused on delivering impact feedback to the hands using wearable or handheld devices [2, 4]. However, these devices typically require users to hold or wear cumbersome hardware. Furthermore, simulating external forces (i.e., force acting on the body from the environment) remains a significant implementation challenge [1].

This study focuses on the transmission path of impact reaction forces. The reaction force generated during impact (e.g., hitting a ball with a racket) propagates through the body, ultimately acting between the feet and the ground. Based on this principle, we propose a method to simulate the action-reaction force between the ground and the body by rapidly rotating a turntable installed beneath the feet. This technique transmits the reaction force from the feet to the entire body, achieving a stronger sense of impact. Furthermore, this approach eliminates the need for reaction force presentation directly to the hands, thereby avoiding the necessity for additional haptic feedback devices attached to the hands.

When striking by swinging the right hand forward, an external force that rotates the body clockwise is generated by the ball (Figure 1 (a)). To counteract this external force, a reaction force that rotates the feet counter-clockwise is generated by the ground at the feet-ground contact point, maintaining the body's equilibrium. Therefore, to express the impact sensation using a turntable, it is necessary to apply a counter-clockwise rotational acceleration to the feet. Furthermore, it is necessary to return the turntable to its initial posture to prevent the body from continuing to rotate.

Based on this physical property, we hypothesize that the impact sensation is most strongly perceived when counter-clockwise

(CCW) angular acceleration is presented to the feet. To effectively present this reaction force, we adopted the "CCW/CW" method, where the turntable is first rotated counter-clockwise (CCW) and then immediately reversed to rotate clockwise (CW). This approach aims to emphasize the initial counter-clockwise acceleration by exploiting the characteristic that people clearly perceive the first strong stimulus.

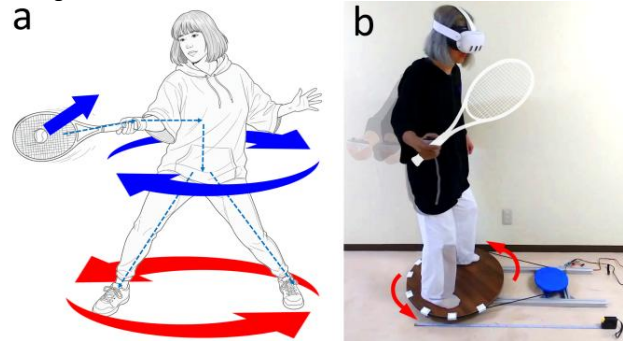


Figure 1 (a) Direction of force applied to the body during impact, (b) Turntable device for impact sensation presentation.

2 TURNTABLE DEVICE

To evaluate the effectiveness of the proposed method, we developed a device to control the rotation of the turntable (Figure 1 (b)). The device consists of an RC servo motor (DS51150, DIYmall), a timing belt, and a 60-cm-diameter turntable. This RC servo motor is controlled using position control via PWM waveforms by a microcontroller (Raspberry Pi Pico W), which receives serial communication data from a PC and rotates the motor to the specified angle.

The presentation of impact sensation is achieved by rotating the turntable—on which the participant is standing—in one direction and then reversing the rotation. In response to a strike, a command for the turntable to rotate is issued first, held for 0.1 seconds, and then a command to rotate in the reverse direction is issued to restore the initial position. The rotation parameters were set by the authors to a degree sufficient for the reaction force to be perceived and were consistent across all participants. Measurement of the rotation angle showed that the maximum rotation angle when a person weighing 48 kg stood on the device was approximately 4 degrees, and the delay from the command to the onset of rotation was less than 0.1 seconds.

3 EVALUATION EXPERIMENT

3.1 Experimental Setup

Seven participants aged 21 to 24 (4 males, 3 females; 6 right-handed) participated in the study. Each participant wore a Meta Quest 3 headset, held a controller in their right hand, and stood on the turntable. The VR viewpoint presented a scene where a ball, launched from the background, collided with a racket held in the participant's right hand.

*†‡ e-mail: {shimada, mizoguchi, kajimoto}@kaji-lab.jp

The user evaluation consisted of two experiments examining the effect of the proposed rotational stimulation on impact sensation. Participants performed eight strikes per condition and then evaluated four items—"Impact Sensation," "Sense of Presence," "Sense of Agency," and "Enjoyment"—using a 7-point Likert scale. The order of experimental conditions was pseudo-randomized across participants to eliminate order effects. All experiments were approved by the ethics board of the authors' institute.

3.1.1 Experiment 1: Turntable Rotation Direction

The objective of this experiment was to determine the optimal turntable rotation direction for generating an impact sensation. The experimental conditions consisted of a single factor: the rotation direction of the turntable. There were three conditions: None (no rotation), CW/CCW (clockwise immediately followed by counter-clockwise), and CCW/CW (counter-clockwise immediately followed by clockwise).

3.1.2 Experiment 2: Combination with Handheld Vibration

The objective of this experiment was to examine the effect on impact sensation when combining conventional controller vibration with turntable rotation. The experimental conditions consisted of two factors: the presence or absence of controller vibration and turntable rotation. The controller vibration was presented using the `OVRInput.SetControllerVibration()` function of Meta XR Core SDK v76 for 0.1 s, with both the frequency and amplitude set to their maximum value of 1. A single vibration pattern was used, providing continuous vibration at the specified intensity. These resulted in four conditions: None (no stimulation), Vib (vibration only), Turn (rotation only), and Vib+Turn (combination of vibration and rotation).

3.2 Results and Discussion

The experimental results were analyzed using multiple comparisons with ART-ANOVA and Bonferroni correction.

3.2.1 Experiment 1

As shown in Figure 2, the CW/CCW condition showed a significant improvement compared to the None condition only in "Enjoyment" ($p=0.0090$). In contrast, the CCW/CW condition showed significant improvements over the None condition in "Impact Sensation" ($p=0.0060$), "Sense of Presence" ($p=0.0029$), and "Enjoyment" ($p=0.0055$). Thus, the hypothesis that a stronger impact sensation is perceived by presenting counter-clockwise (CCW) angular acceleration first was supported. However, it remains unclear which of the two acceleration phases—the initial onset or the subsequent directional change—primarily contributes to the impact sensation. Further investigation is required to decouple these effects.

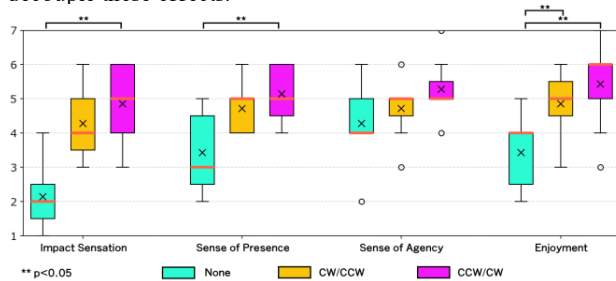


Figure 2 Results of experiment for optimal rotation direction.

3.2.2 Experiment 2

As shown in Figure 3, the Vib+Turn condition showed significant improvements in "Impact Sensation" ($p=0.0392$), "Sense of

Presence" ($p=0.0319$), and "Sense of Agency" ($p=0.0026$) compared to the Vib condition. This demonstrates that the foot rotation stimulus, when combined with handheld vibration, further enhances the impact sensation.

Furthermore, the Turn condition showed significant improvements in all evaluation items compared to the None condition: "Impact Sensation" ($p=0.0039$), "Sense of Presence" ($p=0.0186$), "Sense of Agency" ($p=0.0494$), and "Enjoyment" ($p=0.0038$). Thus, the effectiveness of the foot rotation stimulus alone was also confirmed.

Additionally, the Turn condition showed a significant improvement in "Sense of Agency" ($p=0.0494$) compared to the Vib condition. However, the fact that the scores for the Vib condition were lower than those for the Turn condition may suggest that the intensity of the handheld vibration may have been insufficient. Future work must involve a detailed investigation into how changes in parameters, such as angular velocity and rotation angle, influence the impact sensation. Moreover, it is necessary to clarify the optimal intensity balance with handheld vibration to achieve a more accurate reproduction of the impact sensation.

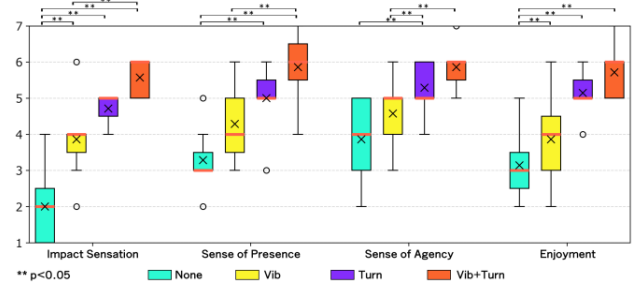


Figure 3 Results of experiment for combined effect of handheld vibration and foot rotation.

4 CONCLUSION

In this study, we proposed a method for presenting impact sensation using a floor-mounted turntable, based on the physical principle that the reaction force during impact travels through the body and acts between the feet and the ground. We hypothesized that for a right-handed strike, a reaction force that rotates the feet counter-clockwise (CCW) is appropriate for the expression of the impact.

User evaluation results showed that the CCW/CW condition significantly improved Impact Sensation, Sense of Presence, and Enjoyment compared to the no-rotation condition, thereby supporting our hypothesis. Furthermore, the results demonstrated that this method, when combined with conventional handheld vibration, serves as an effective means of further enhancing both Impact Sensation and Sense of Presence.

REFERENCES

- [1] C.-A. Fan, et al. SpinShot: Optimizing Both Physical and Perceived Force Feedback of Flywheel-Based, Directional Impact Handheld Devices. In Proc. UIST, ACM, New York, NY, USA, 138:1-15 Oct. 2024. doi:10.1145/3654777.3676433
- [2] J. Lee and S. Choi. Multimodal haptic feedback for virtual collisions combining vibrotactile and electrical muscle stimulation. IEEE Trans. Haptics, 17(1):33-38, 2024. doi:10.1109/TOH.2024.3354268
- [3] Y. Shi and G. Shen. Haptic Sensing and Feedback Techniques toward Virtual Reality. Research, 7:0333, 2024. doi: 10.34133/research.0333
- [4] C.-Y. Tsai, I.-L. Tsai, C.-J. Lai, D. Chow, L. Wei, L.-P. Cheng, and M. Y. Chen. AirRacket: Perceptual Design of Ungrounded, Directional Force Feedback to Improve Virtual Racket Sports Experiences. In Proc. CHI, ACM, New York, NY, USA, 185:1-15 Apr. 2022. doi:10.1145/3491102.3502034